

SENTINEL-GNSS — Individual Poster

Field Data Collection, Cross-City Generalisation Testing, and Results Figures

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My Role

I covered field operations and validation:

- Campus route planning for 5 scenarios
- Led all 34 data collection runs (272 minutes total)
- Cross-geography generalisation tests (UrbanNav HK, Oxford RobotCar)
- Cross-city RF baseline (the 0.15 DEGRADED F1 that makes the paper)
- Generalisation analysis and paragraph writing
- 13 publication-quality figures in Matplotlib/PDF
- Results table formatting for Paper Section 5

Campus Data Collection

Route design — walked the campus to identify:

- Scenario A** (blockage): Library north entrance, 100m open approach
- Scenario B** (urban canyon): Student Centre east corridor, 300m
- Scenario C** (canopy): Dormitory avenue, dense tree cover
- Scenario D** (open sky): East sports ground, 360° horizon
- Scenario E** (gradual): 3 building faces, 5 reps each

Collection protocol: Start 2 min before moving · log blockage times · Joshua verifies on-site.

Scenario	Runs	DEG epochs	CLEAN epochs
A	10	820	1,240
B	5	480	380
C	3	260	2,100
D	1	0	1,920
E	15	1,140	2,350
Total	34	2,700	7,990

Cross-Geography Generalisation Testing

I applied the trained SENTINEL model (**no retraining**) to:

- UrbanNav HK Medium, Deep, Harsh, Tunnel
- Oxford RobotCar GNSS
- UrbanNav Tokyo** (held-out test city)

Critical methodology: Used the *same* normalisation parameters from Beihang training data — never refit to evaluation city. Re-fitting would inflate performance by 5–10%.

The Central Cross-City Result

Model	Tokyo Macro-F1	Tokyo DEG F1
SENTINEL (DL)	0.649	0.753
RandomForest	0.618	0.148

5× gap in safety-critical DEGRADED F1.

I ran the RF evaluation myself to confirm it was not an artifact. The RF memorises Beihang-specific C/N₀ thresholds. Tokyo’s Shinjuku district has different mean C/N₀ (taller buildings, different satellite elevation cutoffs) → RF mis-classifies systematically.

Surprising finding: Tokyo DEGRADED F1 (0.753) > UrbanNav HK Deep (0.687). Shinjuku’s regular skyscraper geometry creates cleaner NLOS transitions than Hong Kong’s irregular canyons. I wrote this interpretation.

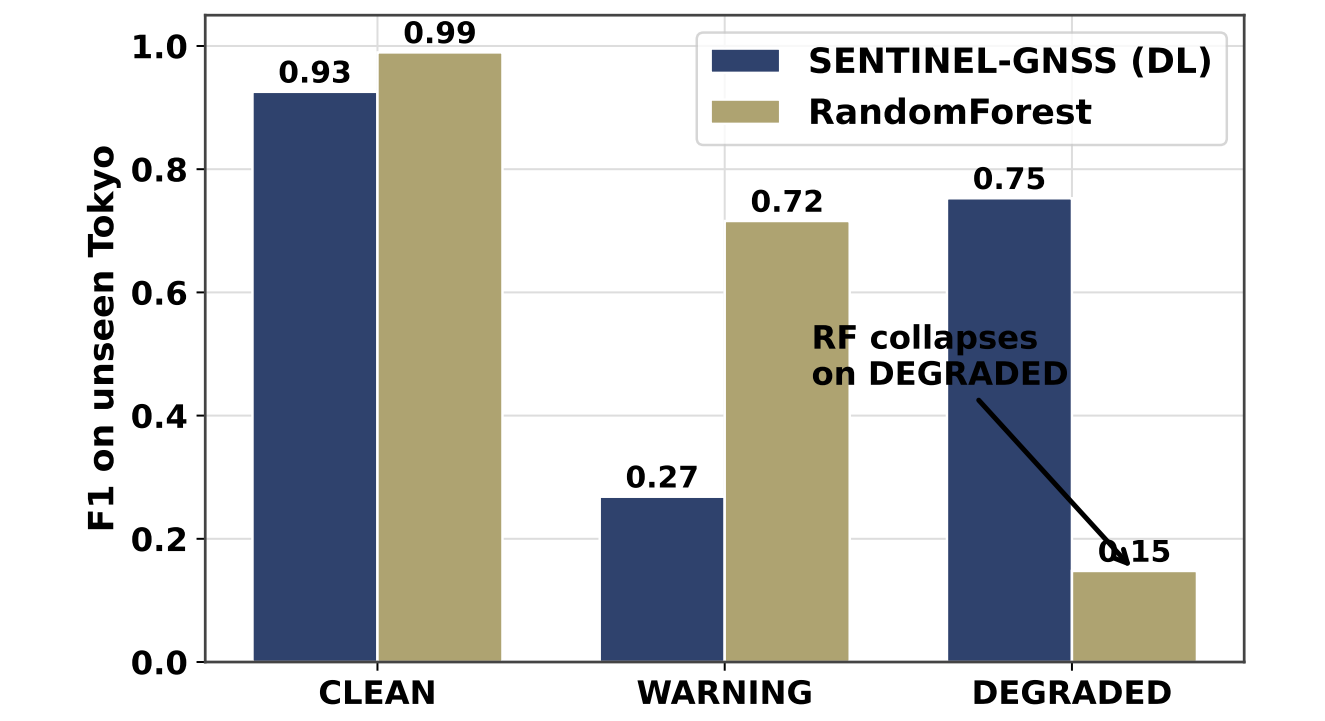


Figure 1: DEGRADED F1 cross-city: DL transfers, RF collapses.

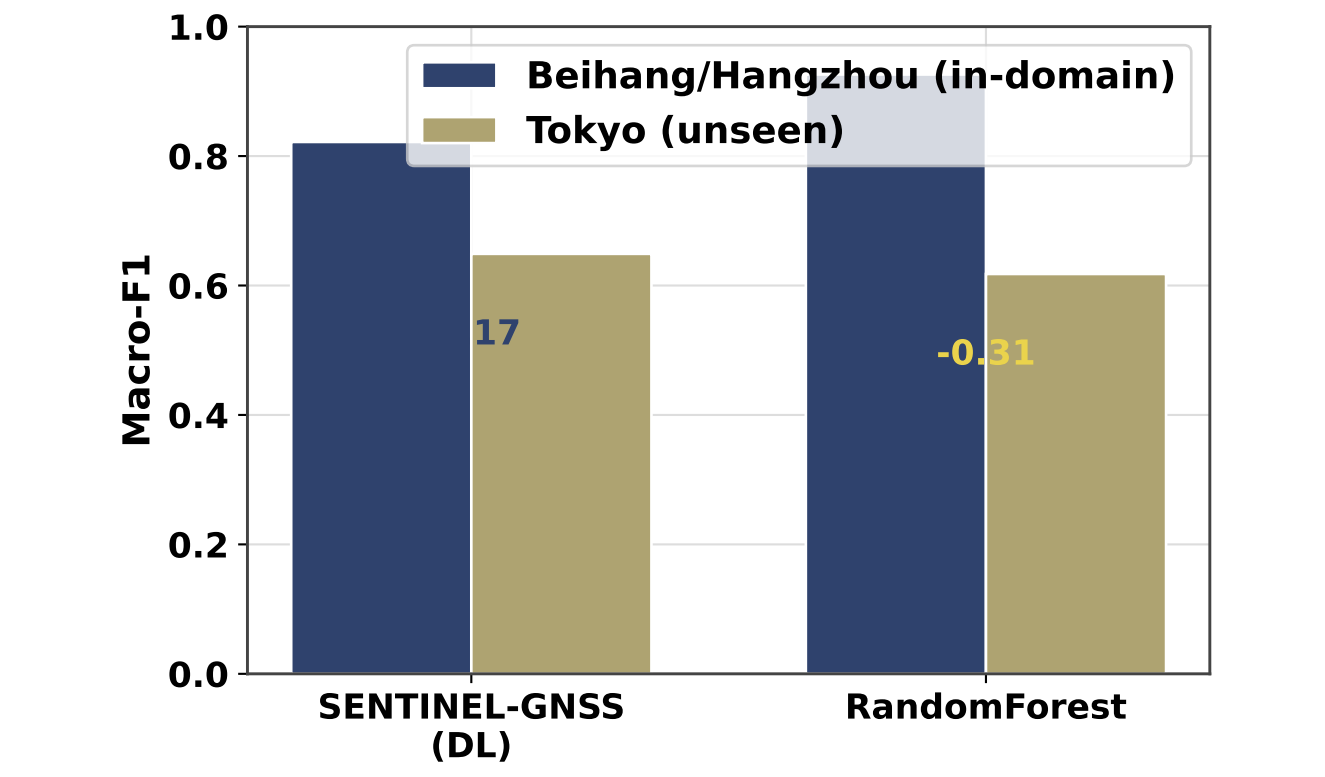


Figure 2: Cross-city Macro-F1 gap. DL gap (−0.17) vs RF (−0.31).

Publication Figures I Produced

Figure	Content
multi_horizon_comparison	Macro-F1 all horizons
confusion_matrices_test	3-panel confusion matrices
pr_curves_test	Precision-recall curves
roc_curves_test	ROC with AUC
lead_time_histogram	Advance warning seconds
learning_curves	Train/val F1 vs epoch
attention_heatmap_*	Transformer attention
feature_saliency_*	Permutation importance

Style: Beihang palette · 12pt minimum axis labels · 300 dpi PNG + PDF vector

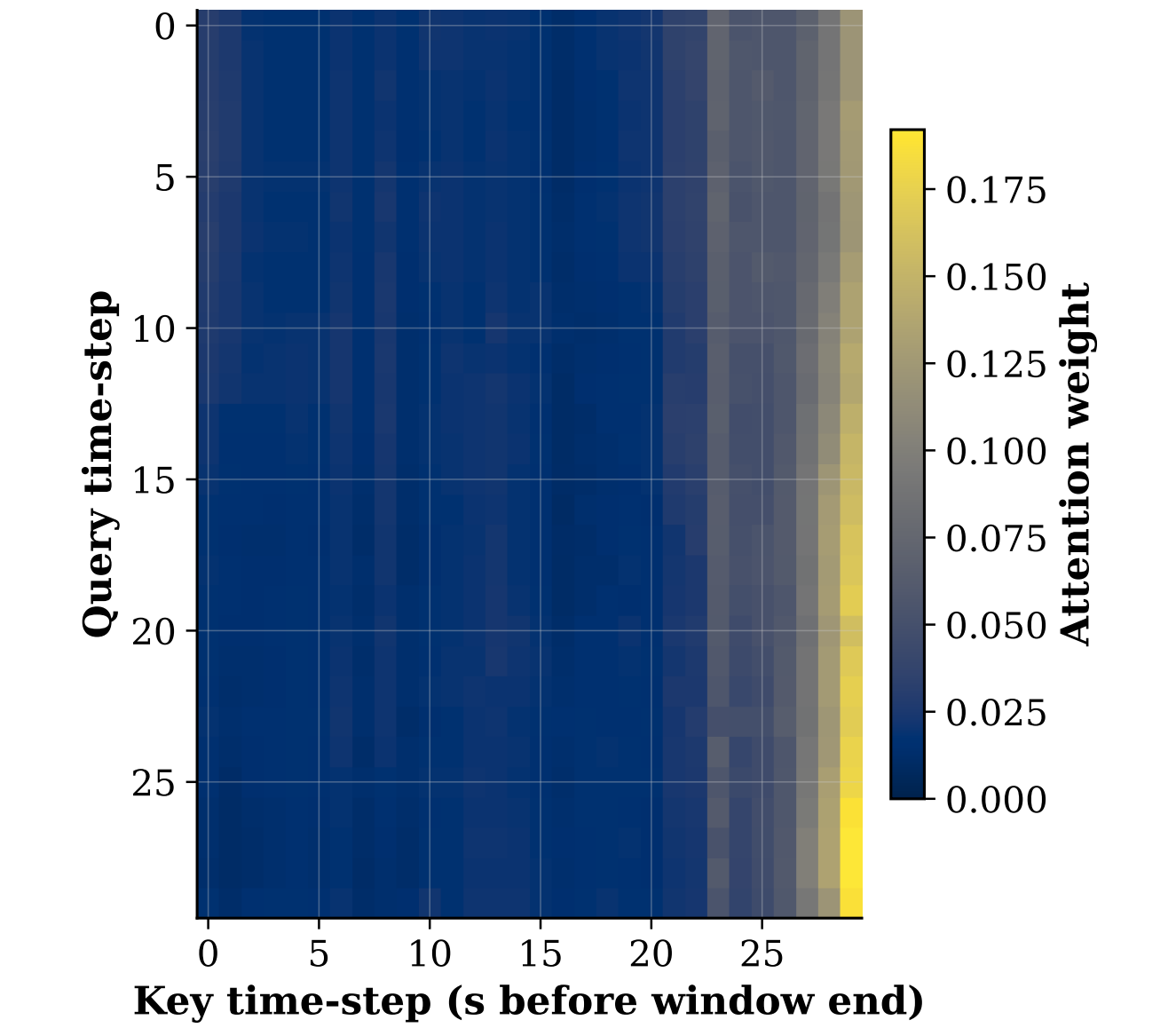


Figure 3: Transformer attention for DEGRADED prediction: strongest attention on the 5–15 epochs before the prediction window.

Key Lesson

Cross-city evaluation is the decisive test, not in-domain leaderboard rank. A classifier that achieves 0.93 Macro-F1 in-domain but collapses to 0.15 DEGRADED F1 on an unseen city is not deployable. The 5× gap I measured is what makes SENTINEL’s contribution real.

Collection lesson: A live SNR display during collection would have saved 4 discarded runs. GNSS signal health should be monitored in real time during field data collection, not discovered post-processing.